

WORLD MODELS ARE BETTER TIME-SERIES LEARNERS

FORECAST-ALIGNED PRETRAINING OF A VIDEO BACKBONE

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ABSTRACT

Rendering a time series as an image and forecasting it by inpainting is the recipe behind the strongest published zero-shot forecasters, all of which inpaint with an ImageNet autoencoder. A video backbone, pretrained on how signals evolve rather than on how they look, ought to be the better prior. We find that it is, but only once the transfer objective is changed. Continuing VideoMAE’s masked-pixel pretraining on a corpus of rendered series improves its own held-out loss monotonically while making forecasts monotonically worse over a $35\times$ range of training budget. The two never meet, because the read-out that converts pixels back into numbers sits outside the loss. We put it inside: the backbone is trained as a world model over rendered series, observing past frames, predicting future ones, and scored by a differentiable read-out that decodes those frames into standardised values. The budget curve inverts. With period-aligned rendering, frame-scale augmentation and 32-frame clips, a single checkpoint under a single inference rule attains 0.285 mean MSE across the six standard long-sequence benchmarks, ahead of VisionTS++ (0.289) and VisionTS (0.315), using 1/13 of VisionTS++’s continual-pretraining steps and 0.3% of its data. Swapping only the initialisation, between image, video and none at all, shows that what transfers is a prior over dynamics, not capacity and not any particular video objective.

1 INTRODUCTION

A time series can be drawn. Once drawn, an image model can finish the drawing, and the finished drawing can be read back as numbers. VisionTS (Chen et al., 2025) showed that this detour beats most purpose-built forecasters in the zero-shot setting, and VisionTS++ (Shen et al., 2025b) showed that continuing the image model’s pretraining on a large corpus of rendered series makes it the strongest zero-shot forecaster published. Both use an ImageNet masked autoencoder (He et al., 2022). Neither has ever seen anything move.

Video backbones are the natural next candidate. They are pretrained on how signals evolve, which is what forecasting asks about, and the recent ones are explicitly built as world models: V-JEPA 2 (Assran et al., 2025) predicts the latent state of the future from the past and is used for planning. If a prior over dynamics is worth anything for forecasting, a video backbone should be where it lives.

We started from that premise and it failed, in a way that turned out to be the interesting part of the story. Taking VideoMAE (Tong et al., 2022) and continuing its native masked-pixel objective on a 654M-observation corpus of rendered series does improve the pretraining objective — held-out pixel loss falls from 0.0366 at 5k steps to 0.0312 at 176k — while downstream forecast error rises over exactly the same range, from 0.339 to 0.381 (Section 4.4). This is not noise and not overfitting: the training loss stays flat, and the degradation is largest at the longest horizons, where one benchmark cell moves from 0.429 to 0.617. More pretraining on more data, with the objective the backbone was designed around, makes the forecaster worse every time we spend more compute on it.

The reason is that the native objective never asks the question we care about. Masked pixel reconstruction is bidirectional and interpolative: a token is recovered from neighbours on both sides, and the model is free to spend capacity on grid lines, anti-aliasing and fill boundaries, none of which carry value information. The mapping from pixels back to numbers — the step that actually determines forecast error — is applied only at test time and is never trained against. Two models can

differ by 8% in gray-level error and by 12% in the opposite direction in forecast error, because the losses measure different things.

Our fix is to make the pretraining task the forecasting task. We train the backbone as a world model over rendered series: the last frames of a clip are masked, the model predicts them from the visible past, and a differentiable soft row-count decodes the predicted frames back into values so that the training loss includes an error term in standardised value units. Nothing about the backbone changes, only what it is asked to be right about. Around this we place three choices that matter at the level of a percent or two each: rendering one period per frame so that vertical position encodes value and horizontal position encodes phase, sampling the periods-per-frame scale during training to match the multi-scale inference rule, and using 32-frame clips instead of 16.

The budget curve inverts. Under the aligned objective, five-dataset mean MSE improves monotonically with training — 0.250 at 5k steps, 0.245 at 20k, 0.242 at 60k — where the pixel objective degraded monotonically over the same range. The final model reaches a six-dataset mean MSE of 0.285 on the standard long-sequence forecasting protocol, ahead of VisionTS++ large (0.289) and base (0.291), Moirai base (0.311) and VisionTS (0.315), at roughly 1/13 of VisionTS++’s continual-pretraining steps and 0.3% of its data. Against the only baseline that publishes per-horizon numbers on this protocol, we win 18 of 24 cells.

A matched-size study then asks what the video backbone was contributing in the first place. Holding the recipe, corpus, budget and inference rule fixed and swapping only the initialisation, an ImageNet MAE-B reaches 0.264 at $H=96$, VideoMAE-B 0.250, V-JEPA 2.1-B 0.252, and the same V-JEPA architecture initialised randomly 0.280 — worse than an image backbone $3.5\times$ smaller. The two video objectives, one reconstructive and one predictive, are indistinguishable from each other while both beat the image backbone. What transfers is a prior over dynamics; the particular objective that produced it is not what matters, and capacity does not explain it.

Contributions.

- A diagnosis: on rendered time series, masked-pixel continual pretraining and forecasting accuracy move in opposite directions across a $35\times$ range of training budget (Section 4.4).
- A forecast-aligned world-model objective that removes the mismatch by decoding predicted frames back to values with a differentiable read-out and training on the value-space error (Section 3). It accounts for the entire gain in our ablation, and the gap grows with horizon.
- State of the art on the standard zero-shot long-sequence benchmark at a fraction of the pretraining budget: 0.285 mean MSE with one checkpoint and one inference rule for all 24 cells (Section 4.2).
- Evidence that the transferable ingredient is a dynamics prior rather than a specific video objective or model capacity, from a matched-size backbone comparison with a random-init control (Section 4.5).

2 RELATED WORK

Visual backbones as forecasters. VisionTS (Chen et al., 2025) renders a univariate series into a periodicity-aligned image, masks the right-hand portion, and lets a frozen ImageNet masked autoencoder (He et al., 2022) inpaint it; decoding the completion gives a forecast that is competitive zero-shot with dedicated time-series models. VisionTS++ (Shen et al., 2025b) keeps the construction and continues pretraining the backbone on LOTSA (Woo et al., 2024), with colourised multivariate rendering and a multi-quantile head; the continual-pretraining stage is what moves it to the top of the zero-shot table. We share the premise that a visual backbone is a reasonable prior for forecasting, and we change the objective that the continual-pretraining stage optimises. To our knowledge this is the first report that the native masked-reconstruction objective degrades forecasting monotonically as its own loss improves. Adjacent evidence points the same way: SVTime (Shen et al., 2025a) matches VisionTS with a 215k-parameter model carrying three inductive biases, and OccamVTS (Lyu et al., 2026) distils a vision forecaster into 1% of its parameters. If most of the transferred capacity can be removed without cost, most of it was not being used, which is what one expects when the objective that selects the capacity is not the one that scores the output. RoMAE (Zivanovic

et al., 2025) is the complementary case: a masked autoencoder made to work on irregular time series by changing the positional representation it sees rather than the architecture around it.

Time-series foundation models. Moirai (Woo et al., 2024), Chronos (Ansari et al., 2024), Time-MoE (Shi et al., 2025), Timer (Liu et al., 2024), TimesFM (Das et al., 2024) and MOMENT (Goswami et al., 2024) pretrain directly on time-series corpora with tokenisation, quantisation or patching schemes designed for the modality. They are our baselines, and the comparison is deliberately unequal in their favour on data: LOTSA in full is 231B observations, and we use 0.3% of it.

Video models as world models. A world model predicts the future state of an environment from its past and an action (LeCun, 2022; Hafner et al., 2025). V-JEPA (Bardes et al., 2024) and V-JEPA 2 (Assran et al., 2025) instantiate this in video by predicting the latent representation of masked future regions, and are used downstream for planning. VideoMAE (Tong et al., 2022) is not a world model in this sense — it is a bidirectional masked autoencoder whose prior over dynamics is implicit — but the distinction turns out to matter less than the objective it is fine-tuned with. We use both families as backbones and, once the forecast-aligned objective is applied, both become world models over rendered series in the literal sense: past frames in, future frames out, supervised on the decoded values.

Transfer and anchoring. Continual pretraining of a strong backbone on a narrow corpus risks erasing the prior it was chosen for. L2-SP (Li et al., 2018) regularises toward the initial weights instead of toward zero. We use it as an ablation arm and report where it matters (16-frame clips) and where it is inert (32-frame clips), which is itself informative about what the video prior supplies.

3 A WORLD MODEL OVER RENDERED SERIES

The model observes the first $F - h_p$ frames of a rendered clip and predicts the last h_p . Training supervises that prediction in the space where forecasts are scored. Everything else in this section exists to make those two sentences well defined.

3.1 RENDERING: ONE PERIOD PER FRAME

Let $x_{1:L}$ be a univariate series with dominant period P , taken from the sampling frequency as in Chen et al. (2025). We render $L = 16kP$ observations into F frames of 224×224 , where k is the number of periods per frame. Values are standardised with the mean and standard deviation of the *visible context only*, so no test-time statistic leaks into the input, then clipped to $z \in [-3, 3]$ and mapped to a column height

$$h = 0.5 + 0.4 \frac{z}{3} \in [0.1, 0.9]. \quad (1)$$

Column c of a frame is drawn as an area chart: row $\text{round}((1 - h) \cdot 223)$ is the boundary, rows below it are filled with gray level 0.12, rows above hold background 0.92, and horizontal grid lines every 28 rows carry 0.80. Frame f holds periods $[fk, (f + 1)k)$. Vertical position therefore encodes value and horizontal position encodes phase within the period, so a seasonal series becomes a nearly static video and a trending or regime-changing one becomes a moving one. With $F = 32$ and a tubelet size of 2, a clip is 16 temporal by 14×14 spatial tokens.

3.2 MASKING: THE FUTURE, NOT A RANDOM SUBSET

The last $h_p \in \{2, 4\}$ frames are masked; these are the future. With probability 0.3 the leading tubelets are hidden as well, so the model also trains on contexts shorter than the full window, which is the regime it meets at test time on short datasets. To retain some of the original pretraining signal, 30% of batches instead use standard tube masking. The reconstruction loss is computed on the masked future tokens only.

3.3 THE FORECAST-ALIGNED OBJECTIVE

Pixel error is a poor surrogate for forecast error because the decoder that turns pixels into numbers is not in the loop. We put it there. For a predicted frame $\hat{x}$, define the per-pixel soft occupancy

$$s(\hat{x}_{r,c}) = \text{clamp}\left(\frac{0.80 - \hat{x}_{r,c}}{0.80 - 0.12}, 0, 1\right), \quad \hat{h}_c = 1 - \frac{224 - \sum_r s(\hat{x}_{r,c})}{223}, \quad (2)$$

which counts, differentiably, how much of column c is filled. The thresholds are the fill level and the grid-line level from Section 3.1, so grid lines contribute a bounded, constant offset rather than gradient noise. Inverting Equation 1 gives $z = (h - 0.5) \cdot 3/0.4$, and the training objective is

$$\mathcal{L} = \underbrace{\text{MSE}(\hat{x}, x)}_{\text{pixel}} + \lambda \left(\frac{3}{0.4}\right)^2 \underbrace{\text{MSE}(\hat{h}_c, h_c)}_{\text{value}}, \quad \lambda = 0.1, \quad (3)$$

where the constant $(3/0.4)^2$ converts height units into standardised value units, so the second term is an MSE in exactly the units the benchmark reports. It is computed column-wise and is therefore independent of P , of the dataset, and of the number of periods packed into a frame. The pixel term is kept because it stabilises early training, when the read-out of a blurry prediction is uninformative; $\lambda = 0.1$ was the first value we tried and 0.3 is worse (Appendix D).

This is the whole of the method’s gain. It is also what makes the model a world model over rendered series rather than an inpainter: the quantity being optimised is the future state of the series, in the units of the series.

3.4 FRAME-SCALE AUGMENTATION AND PRIOR ANCHORING

At test time a long horizon is covered by packing more periods into each frame, so the model must work at several values of k . During training we sample $k \in \{1, 2, 4\}$ with probability 0.5/0.3/0.2, matching the inference rule below. Independently, we can decay weights toward their initial values θ_0 rather than toward zero, $p \leftarrow p - \eta \lambda_{\text{sp}}(p - \theta_0)$ with $\lambda_{\text{sp}} = 0.05$ and weight decay off (Li et al., 2018). This matters at 16-frame clips and is inert at 32 (Table 3), which we read as evidence that one thing the video prior supplies is long-range temporal context — something a longer window supplies directly.

3.5 INFERENCE

One checkpoint and one rule produce every number we report. Given period P and horizon H : if $H \leq 4P$, a single pass with $h_p = 4$, averaged over the scales k with $kPs \leq 96$; otherwise $k = \lceil H/4P \rceil$ periods per frame, followed by $\lceil k/(96/P) \rceil$ autoregressive rollout passes when $P < 96$ and a single direct pass when $P \geq 96$. There is no per-dataset tuning, no ensembling over checkpoints and no blending with statistical baselines anywhere in the paper.

3.6 BACKBONE, OPTIMISATION AND CORPUS

The main model continues VideoMAE-B pretrained on Kinetics-400 (Tong et al., 2022) with `norm_pix_loss` disabled. VideoMAE’s position embeddings are sinusoidal, so moving from 16 to 32 frames rebuilds the tables and loads the pretrained weights unchanged. We use AdamW (Loshchilov & Hutter, 2019) ($\beta = 0.9/0.95$, learning rate 10^{-4} with cosine decay and 500 warm-up steps), batch 32 as micro-batch 8 with 4-step accumulation, and bf16; 60k steps take about 20 hours on one A40.

The corpus is 98 LOTSA subsets (Woo et al., 2024): 184,013 series and 654.2M observations, with 20% synthetic series mixed in and subsets sampled in proportion to the square root of their size. None of the six evaluation datasets appears in it. We note one constraint this creates: the corpus contains Monash `traffic_hourly` (Godaheva et al., 2021), which is the same data as the LSF traffic benchmark, so traffic is excluded from our evaluation rather than reported as a zero-shot result.

Table 1: Zero-shot long-sequence forecasting. Each entry averages the four horizons $H \in \{96, 192, 336, 720\}$; MSE and MAE are computed on standardised values. Our model uses one checkpoint and one inference rule for all 24 cells. Baselines are quoted from Shen et al. (2025b) and Chen et al. (2025); ‘-’ means the source does not report that dataset. **Bold**: best. Underlined: second best. The last column counts first places over the twelve dataset columns.

Model	ETTm1		ETTm2		ETTh1		ETTh2		ECL		Weather		Avg		1 st
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	
Ours	0.352	0.369	0.241	<u>0.302</u>	0.404	<u>0.416</u>	0.335	0.376	0.163	0.254	0.215	0.252	0.285	0.328	6
VisionTS++ (ViT-L)	<u>0.354</u>	0.369	<u>0.244</u>	0.298	0.403	0.418	0.327	0.365	<u>0.181</u>	<u>0.264</u>	0.226	<u>0.243</u>	<u>0.289</u>	0.326	4
VisionTS++ (ViT-B)	<u>0.360</u>	<u>0.372</u>	<u>0.244</u>	0.298	0.402	<u>0.416</u>	<u>0.333</u>	<u>0.370</u>	<u>0.184</u>	<u>0.265</u>	<u>0.222</u>	0.241	0.291	<u>0.327</u>	2
VisionTS	0.374	<u>0.372</u>	0.318	0.366	0.390	0.414	<u>0.333</u>	0.375	0.207	0.294	0.269	0.292	0.315	0.352	2
Moirai _S	0.448	0.410	0.272	0.321	0.400	0.424	0.341	0.379	0.233	0.320	0.242	0.267	0.323	0.353	-
Moirai _B	0.382	0.388	0.276	0.320	0.434	0.439	0.346	0.382	0.188	0.274	0.238	0.261	0.311	0.344	-
Moirai _L	0.390	0.389	0.317	0.366	0.510	0.469	0.354	0.377	0.188	0.273	0.260	0.275	0.337	0.358	-
Chronos _S	0.640	0.500	0.310	0.350	0.545	0.472	0.424	0.430	0.220	0.284	0.300	0.318	0.407	0.392	-
Chronos _B	0.646	0.500	0.295	0.338	0.591	0.468	0.406	0.411	0.215	0.279	0.293	0.315	0.408	0.385	-
Chronos _L	0.556	0.465	0.300	0.341	0.589	0.466	0.455	0.427	0.204	0.274	0.279	0.306	0.397	0.380	-
Time-MoE _S	0.394	0.416	0.316	0.361	0.400	0.424	0.367	0.404	-	-	0.266	0.297	-	-	-
Time-MoE _B	0.376	0.406	0.349	0.380	<u>0.394</u>	0.420	0.405	0.415	-	-	0.270	0.300	-	-	-
Timer	0.487	0.457	0.328	0.347	0.444	0.457	0.358	0.407	-	-	0.304	0.331	-	-	-
TimesFM	0.433	0.419	-	-	0.473	0.444	0.392	0.406	-	-	-	-	-	-	-
MOMENT	0.670	0.537	0.316	0.371	0.684	0.566	0.362	0.410	0.765	0.687	0.294	0.326	0.515	0.483	-

4 EXPERIMENTS

4.1 SETUP

We follow the zero-shot long-sequence forecasting protocol of Chen et al. (2025) exactly: Time-Series-Library splits (Wu et al., 2023) (ETT 12/4/4 months, others 70/10/20), a StandardScaler fitted on the training portion, all channels, stride-1 test origins, and horizons $H \in \{96, 192, 336, 720\}$, with MSE and MAE computed on standardised values. Context lengths and periods per dataset follow Chen et al. (2025): ETTh1 2880/24, ETTh2 1728/24, ETTm1 2304/96, ETTm2 4032/96, electricity 2880/24, weather 4032/144. As a check on the harness we ran their released checkpoint through our pipeline and reproduced their numbers to three decimals (ETTh1 at $H=96$: 0.3527/0.3833 against the published 0.353/0.383). Baseline numbers are quoted from Shen et al. (2025b) Table 4 and from Chen et al. (2025).

Ablations use the five datasets that fit a single evaluation budget (ETT $\times$ 4 and weather) at three horizons; the full six-dataset, four-horizon grid is reserved for the main table and for seed replication.

4.2 MAIN RESULTS

Table 1 places the model at the top of the published zero-shot table by mean MSE, 0.285 against 0.289 for VisionTS++ large and 0.291 for its base model. Four of six datasets are won outright. The two losses are ETTh1, which goes to VisionTS (0.390 against our 0.404), and ETTh2, which goes to VisionTS++ large (0.327 against 0.335); both are hourly ETT sets at mid-to-long horizons, where a longer tuned context helps and our window is capped at 32 frames of at most four periods. On mean MAE we tie rather than win, 0.328 against 0.326. The margin on electricity (0.163 against 0.181) and weather (0.215 against 0.222) is where the aggregate gain comes from. Trained for 20k steps instead of 60k — 6.3 GPU-hours — the same recipe already reaches 0.288/0.331, which would still lead the table.

Table 2 breaks the aggregate down, and the head-to-head in Appendix B gives us 18 of VisionTS’s 24 cells. The six losses are all mid-to-long horizons on hourly ETT, and they share a cause with the ETTh1/ETTh2 column losses above: at $H=720$ with $P=24$, covering the horizon requires either eight periods per frame or several rollout passes, and both cost resolution. The wins are large and systematic elsewhere: every electricity cell improves by 18–28% over VisionTS, and weather at $H=96$ by 37%.

Table 2: Our model, per horizon. Same checkpoint and inference rule as Table 1. The head-to-head against VisionTS, the only baseline that publishes per-horizon numbers on this protocol, is in Appendix B: we win 18 of its 24 cells.

Dataset	$H=96$		$H=192$		$H=336$		$H=720$		Avg	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTm1	0.292	0.332	0.332	0.357	0.363	0.377	0.423	0.409	0.352	0.369
ETTm2	0.160	0.245	0.214	0.283	0.264	0.318	0.328	0.362	0.241	0.302
ETTh1	0.352	0.380	0.396	0.408	0.424	0.425	0.446	0.449	0.404	0.416
ETTh2	0.266	0.324	0.333	0.369	0.361	0.395	0.379	0.415	0.335	0.376
ECL	0.128	0.219	0.148	0.239	0.170	0.262	0.207	0.294	0.163	0.254
Weather	0.139	0.183	0.181	0.226	0.232	0.266	0.308	0.331	0.215	0.252
Mean	0.223	0.281	0.267	0.314	0.302	0.341	0.349	0.377	0.285	0.328

Table 3: Ablations. Five-dataset mean MSE (ETT $\times$ 4 and weather), 32-frame clips, identical corpus, recipe and inference rule.

Configuration	$H=96$	$H=336$	$H=720$
Full recipe, 20k steps	0.245	0.330	0.379
– value-space loss (native pixel objective)	0.257	0.347	0.411
– prior anchoring	0.244	0.330	0.378
5k steps	0.250	0.337	0.384
60k steps	0.242	0.329	0.377

4.3 WHAT THE OBJECTIVE BUYS

Removing the value-space term and training the same backbone on the same data with the native pixel objective costs 5% at $H=96$, 5% at $H=336$ and 8% at $H=720$ (Table 3). The term accounts for the entire gain of the recipe, and its value grows with horizon, which is what one expects if the mechanism is the alignment between what is optimised and what is scored: the longer the horizon, the more of the prediction is generated rather than copied. Prior anchoring is inert at 32 frames.

The budget rows matter as much as the ablation rows. Under the aligned objective, error decreases monotonically from 5k to 60k steps. Under the pixel objective it increases monotonically over the same range, which is the subject of the next subsection. Two further attempts to buy accuracy with scale did not work and we report them for completeness: raising the batch size to 128 or 256 is equal or worse at every horizon, and quadrupling the in-domain corpus (adding buildings_900k, LargeST, Q-Traffic and Azure) ties the corpus we use.

4.4 THE OBJECTIVE MISMATCH

Table 4 and Figure 1a are the result that produced the method. Continuing VideoMAE’s own objective on the rendered corpus improves its own held-out loss by 15% while forecast error rises by 12%. Per cell, the 176k model is 5–60% worse than the 20k model, with the damage concentrated at long horizons; ETTh2 at $H=720$ moves from 0.429 to 0.617.

We checked the obvious alternative explanations. The training loss is flat across the range (0.0092 $\rightarrow$ 0.0089), so this is not pixel-space overfitting. Checkpoint key sets, the transformers version, the attention configuration and loader warnings were all ruled out by re-evaluating the same checkpoints under matched software. What does change monotonically is the weight norm: all layers shrink by roughly 30% over the 9 $\times$ longer schedule under AdamW’s decoupled decay, so the backbone drifts steadily away from the initialisation that made it useful while its own objective keeps improving. The aligned objective removes the symptom, and prior anchoring, which attacks the drift directly, helps only in the regime where the window is short (Section 4.3).

Table 4: Masked-pixel continual pretraining, 16-frame clips. The pretraining objective improves monotonically; forecasting degrades monotonically. Downstream is the five-dataset mean MSE over three horizons.

Training steps	5k	20k	60k	176k
Held-out pixel loss	0.0366	0.0333	0.0329	0.0312
Downstream mean MSE	0.339	0.344	–	0.381

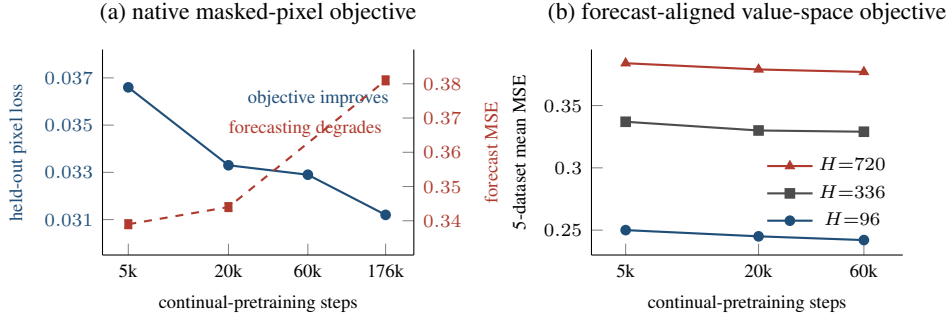


Figure 1: The same backbone and corpus under two objectives. **(a)** VideoMAE’s native masked-pixel objective: its own held-out loss falls monotonically while forecast error rises monotonically over a $35\times$ range of budget. The 60k checkpoint was not evaluated downstream. **(b)** The forecast-aligned value-space objective of Section 3.3: the curve is monotonically improving at every horizon. Panel (a) uses 16-frame clips and panel (b) 32-frame clips; the ablation in Table 3 compares the two objectives at matched clip length.

4.5 WHAT THE BACKBONE CONTRIBUTES

Table 5 holds everything fixed except the weights the model starts from. At matched size, both video backbones beat the image backbone by 3–5% at every horizon, and they are indistinguishable from each other: 0.250 for a reconstructive video model against 0.252 for a predictive one at $H=96$, with the order reversing at longer horizons. The advantage of V-JEPA 2-L (0.245) over V-JEPA 2.1-B is capacity, not objective, and we do not read it as evidence that predictive world models beat reconstructive ones.

The random-init control is the informative row. The same ViT-L architecture, same data, same steps, same everything, reaches 0.280 at $H=96$ — 12.5% worse than its pretrained twin, and worse than an image backbone $3.5\times$ smaller. Per dataset at $H=96$ it gives 0.401/0.282/0.369/0.182/0.166 against 0.357/0.270/0.294/0.162/0.141 pretrained; the gap is 10.8% at $H=336$ and 9.3% at $H=720$. Capacity does not explain the transfer, and neither does the corpus, which all arms share. What is left is the prior over dynamics.

Three caveats on the V-JEPA 2.1-B arm: it is distilled from a ViT-G teacher, its predictor output head was re-initialised because the pretrained head maps into the 1664-dimensional teacher space while our targets are 768-dimensional ViT-B EMA features, and it was pretrained at 384px while we render at 224.

4.6 SEED VARIANCE AND EFFICIENCY

Two seeds of the 20k model agree to within 0.002 on the five-dataset means (0.245/0.330/0.379 and 0.243/0.330/0.378), and on the full six-dataset grid they agree on five of six datasets, with ETTm1 the only visible spread (0.351 against 0.360). Averaging the two seeds in *prediction* space is worth a 0.6% and is never worse on any cell, giving 0.286 against the single-seed 0.288 at 20k steps. We leave it out of Table 1, since it doubles the training cost for a gain smaller than the step from 20k to 60k.

Table 5: Backbone study. Identical recipe, corpus, 5k-step budget, 32-frame clips and inference rule; only the initialisation changes. Five-dataset mean MSE.

Initialisation	Pretraining signal	Params	$H=96$	$H=336$	$H=720$
ImageNet MAE-B	images, no dynamics	87M	0.264	0.350	0.395
VideoMAE-B	video, implicit dynamics	87M	0.250	0.337	0.384
V-JEPA 2.1-B	video, explicit prediction	87M + 23M	0.252	0.335	0.382
V-JEPA 2-L	video, explicit prediction	300M	0.245	0.329	0.379
V-JEPA 2-L, random init	none	300M	0.280	0.369	0.418

Table 6: Pretraining budget. Steps are counted in windows seen; we do not claim a FLOP comparison, since the architectures differ.

	Continual-pretraining windows	Corpus	Wall clock
Ours (60k, main)	$60k \times 32 = 1.9M$	654M observations	≈ 20 h, $1 \times A40$
Ours (20k row)	$20k \times 32 = 0.64M$	same	6.3 h, $1 \times A40$
VisionTS++	$\approx 100k$ steps, large batch	LOTSAs, 231B observations	not reported

Table 6 states the budget comparison in the only units we can measure on both sides: pretraining windows and corpus size. We use roughly 1/13 of VisionTS++’s continual pretraining steps (1/40 for the 20k row) and 0.3% of its data. We deliberately do not phrase this as a compute claim; the models differ in size and our 32-frame clips are about $8\times$ more expensive at inference than 16-frame clips.

5 DISCUSSION

The objective, not the architecture. The same backbone, the same corpus and the same number of steps produce either a forecaster that improves with training or one that degrades with training, depending only on what the loss measures. That is a stronger statement than ”our loss is better”: it says the usual practice of continuing a vision model’s native objective on rendered series is not a neutral default but an active liability at scale, and that the community’s instinct to buy accuracy with more pretraining will be punished here. It also explains why the visual prior found by prior work looks small (Shen et al., 2025a): when the read-out is outside the loop, most of the capacity being transferred is being spent on things the read-out discards.

What a video prior actually supplies. Two independent observations suggest the answer is long-range temporal context rather than anything specific to video semantics. First, at 16-frame clips the video initialisation beats the image initialisation by 3–4%, and at 32 frames the two converge (Appendix C); lengthen the window and part of the prior becomes redundant. Second, prior anchoring is essential at 16 frames and inert at 32. Both point at the same thing: the prior is worth most exactly where the input does not already contain the structure. The matched-size comparison at a short budget (Table 5) is where the video prior is visible, and the random-init control shows it is real rather than an artefact of capacity.

We are careful about what this does not say. It does not say video backbones beat image backbones in general; at 20k steps under the aligned objective the gap closes. It does not say predictive world models beat reconstructive ones; at matched size they tie. The defensible claim is narrower and, we think, more useful: a prior over dynamics transfers to forecasting, several pretraining objectives produce one, and the objective used during transfer decides whether any of it survives.

Why the value-space term grows with horizon. At short horizons a correct completion is largely determined by the visible context, and pixel error and value error rank candidate completions similarly. As the horizon grows, more of the prediction is generated rather than interpolated, and the two metrics diverge: a prediction can be smooth, plausible and pixel-cheap while being wrong by a

constant offset in value. The ablation matches this reading, with the gap growing from 5% at $H=96$ to 8% at $H=720$.

Limitations. Mid-to-long horizons on the hourly ETT sets remain behind VisionTS and VisionTS++, and the cause is structural: our context is capped at 32 frames of at most four periods, so at $H=720$ with $P=24$ we trade resolution for coverage. Mean MAE ties rather than beats the strongest baseline (0.328 against 0.326), which is worth stating plainly since we lead with MSE. The evaluation is the six-dataset LSF protocol only; GIFT-Eval (Aksu et al., 2024) and Monash (Godahewa et al., 2021) results are not yet in. Our budget comparison is in pretraining windows and observations, not FLOPs. The main result uses one backbone family, with the V-JEPA arms serving as 5k-step controls rather than as full models. Finally, the pretraining corpus contains Monash `traffic.hourly`, so the traffic benchmark cannot be added without rebuilding the corpus, and we exclude it rather than report a contaminated number.

6 CONCLUSION

Video backbones do help time-series forecasting, but not through the objective they were trained with. Continuing masked-pixel pretraining on rendered series improves that objective and degrades forecasting at the same time, monotonically, over a $35\times$ range of budget. Training the same backbone as a world model over rendered series — past frames in, future frames out, supervised through a differentiable read-out in value space — inverts the curve and produces the best mean MSE in the published zero-shot table with a small fraction of the pretraining budget of the models it displaces. A matched-size comparison with a random-init control locates the transferable ingredient in a prior over dynamics rather than in capacity or in any one video objective.

The practical reading is that the rendering detour is not the interesting design decision; the loop between prediction and read-out is. Anything that turns a visual model into a forecaster eventually converts pixels into numbers, and our results suggest that conversion belongs inside the training objective rather than after it. The obvious next steps are a second benchmark, a longer effective context to recover the hourly ETT cells, and porting the value-space loss into the image-model pipeline it was diagnosed on.

REPRODUCIBILITY STATEMENT

Every number in this paper comes from one publicly documented pipeline. The renderer, the value-space read-out, the trainer and all command-line flags used for each arm are described in Section 3 and Appendix A; the evaluation follows the protocol of Chen et al. (2025) and was checked against their released checkpoint to three decimals (Section 4.1). One checkpoint and one inference rule produce all 24 evaluation cells: no per-dataset tuning, no checkpoint ensembling, no blending with statistical baselines. The pretraining corpus is built from public LOTSA subsets (Woo et al., 2024) with a documented subset list, and none of the six evaluation datasets appears in it. Code, corpus manifest and checkpoints will be released.

ETHICS STATEMENT

The work uses public forecasting benchmarks and a public pretraining corpus; no human subjects, no personally identifying data. Electricity and traffic-like series describe aggregate consumption at substation level. The main risk we see is the usual one for forecasting models sold as general purpose: the reported zero-shot accuracy is measured on six datasets in two domains and should not be read as a guarantee on safety-critical deployments.

USE OF LARGE LANGUAGE MODELS

Large language models were used for copy-editing and for scripting routine experiment orchestration. All experimental design, analysis and claims are the authors’; every number reported here was produced by the pipeline described above.

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A IMPLEMENTATION DETAILS

Rendering. Frames are 224×224 grayscale, replicated to three channels. Fill level 0.12, background 0.92, grid lines every 28 rows at 0.80. Values are standardised with the statistics of the visible context, clipped to $z \in [-3, 3]$ and mapped to height by Equation 1. A clip holds $16kP$ observations in F frames; tubelet size is 2, giving $16 \times 14 \times 14 = 3136$ tokens at $F = 32$.

Training. VideoMAE-B initialised from Kinetics-400, `norm_pix_loss` off, AdamW ($\beta_1=0.9$, $\beta_2=0.95$), learning rate 10^{-4} with cosine decay and 500 warm-up steps, weight decay 0.05 (or 0 with prior anchoring at $\lambda_{sp} = 0.05$), bf16, micro-batch 8 with 4-step gradient accumulation for an effective batch of 32. Masked future length h_p is sampled from $\{2, 4\}$; with probability 0.3 the leading tubelets are additionally hidden; 30% of batches use tube masking instead. Frame scale $k \in \{1, 2, 4\}$ with probability 0.5/0.3/0.2. Learning rate 2×10^{-4} is worse and 5×10^{-5} ties 10^{-4} .

Corpus. 98 LOTSA subsets, 184,013 series, 654.2M observations, with 20% synthetic series (sinusoids with trend, noise and regime changes) mixed in. Subsets are sampled with probability proportional to the square root of their observation count. We verified that none of ETTh1, ETTh2, ETTm1, ETTm2, electricity or weather is present; the nearest same-domain entries are Monash weather (daily Australian stations, not the Jena 10-minute set) and `traffic.hourly`, which is why traffic is excluded from the evaluation.

Evaluation. `--mode auto` as described in Section 3.5. All 24 cells use the same checkpoint. Multi-channel datasets are forecast channel by channel. Reported MSE and MAE are computed on standardised values over stride-1 test origins.

B PER-CELL COMPARISON AGAINST VISIONTS

C INITIALISATION AT TWO CLIP LENGTHS

At 16 frames the video prior is worth 3–4%; at 32 frames the two initialisations tie. Read together with the anchoring ablation (inert at 32 frames, essential at 16), this suggests that part of what video pretraining supplies is long-range temporal context, which a longer window supplies directly. The matched-size comparison in Table 5 therefore uses the 5k-step budget, where the prior is still doing work.

Table 7: MSE per dataset and horizon against VisionTS, the only baseline that publishes per-horizon numbers on this protocol. **Bold**: better of the two. We win 18 of 24 cells; the six losses are all mid-to-long horizons on the hourly ETT sets.

Dataset	$H=96$		$H=192$		$H=336$		$H=720$	
	Ours	VisionTS	Ours	VisionTS	Ours	VisionTS	Ours	VisionTS
ETTm1	0.292	0.341	0.332	0.360	0.363	0.377	0.423	0.416
ETTm2	0.160	0.228	0.214	0.262	0.264	0.293	0.328	0.343
ETTh1	0.352	0.353	0.396	0.392	0.424	0.407	0.446	0.406
ETTh2	0.266	0.271	0.333	0.328	0.361	0.345	0.379	0.388
ECL	0.128	0.177	0.148	0.188	0.170	0.207	0.207	0.256
Weather	0.139	0.220	0.181	0.244	0.232	0.280	0.308	0.330

Table 8: Video against image initialisation at two clip lengths, two seeds each where available. Five-dataset mean MSE at $H=96/336/720$.

Clip length	Video init		Image init	
16 frames	0.257/0.350/0.383,	0.256/0.348/0.384	0.268/0.359/0.400,	0.265/0.353/0.400
32 frames	0.245/0.330/0.379		0.244/0.329/0.373,	0.245/0.330/0.373

D ALTERNATIVES THAT DID NOT WORK

We list these so that the obvious questions have answers, not because any of them is a finding.

Table 9: Attempts that did not improve on the recipe in Section 3.

Attempt	Outcome
9× budget under the pixel objective (176k steps)	5–60% worse per cell
Larger batch (128, 256 via accumulation)	equal or worse at every horizon
4× more in-domain data (buildings_900k, LargeST, Q-Traffic, Azure)	ties the corpus we use
Longer masked future (8 of 16 frames)	slightly worse
Value-loss weight 0.3 instead of 0.1	worse
Three-checkpoint ensemble	no gain
Blending with a zero-parameter statistical prior	validation selection does not transfer
Weight-space averaging of two seeds	better on ETT, 16% worse on electricity, net worse
Cross-corpus weight averaging	worse than either parent

Prediction-space averaging of two seeds is the one combination that helps, by 0.6% and never worse on any cell; it appears as a secondary row in Table 1.